General questions

Welcome to the Survey

This survey, titled "*Real Faults in Model Context Protocol (MCP) Servers*", is conducted as part of ongoing research on Model Context Protocols (MCP) at the SWAT Lab, Polytechnique Montréal, Canada. The survey takes approximately 10 minutes to complete and focuses on understanding the relevance and severity of faults in MCP server development. It is important to note that the survey must be completed in a single session. Responses cannot be saved partway through and resumed at a later time. Participation is voluntary and anonymous; no names, email addresses, or IP information will be collected. You are also encouraged to share the survey with colleagues or students who may be interested in contributing. The aggregated results of this study will be made publicly available on **arXiv.org.**

Consent

1. I have read and understood the attached documentation, which describes the nature and

the process of the research project, as well as the risks and inconveniences that this project may incur.
2. I understand that as a participant in this research project, I do not renounce any of my rights, nor do I waive researcher's legal responsibilities.

- ○ I accept
- ○ I refuse

Your email address (optional)

What is your current job?
Developer, Researcher, ... (for students, please indicate the degree: PhD, Master, etc.)

Which programming languages have you used to develop MCP servers?

- Python
- Kotlin
- C #
- Ruby
- Rust
- TypeScript
- Go
- PHP
- Java
- Swift

Please specify any other programming languages not listed above

Have you ever developed or contributed to the development of an MCP server?

- ○ Yes, MCP core
- ○ Yes, non-MCP core
- ○ No

If you answered "yes" to the previous question, could you please share links to the MCP repositories you have developed or contributed to (if you are comfortable sharing them)?

Server Setting

Our study has identified 4 high-level common categories of faults in MCP development including *1) Server setting*, *2) Server/host configuration*, *3) Server/tool configuration*, and *4) Documentation*.

Have you ever encountered the following faults related to **'MCP server setting'**?
Faults with server configuration and required setup for proper MCP server operation

	Encountering the fault		How severe was the fault			How much effort was required to handle the fault		
	Yes	No	Minor	Major	Critical	Low	Medium	High
Server Deployment Faults during server packaging, distribution, or deployment, such as build or installation failures	○	○	○	○	○	○	○	○
Server Network Configuration Faults with network setup, connectivity, firewall/proxy settings, or accessing remote servers	○	○	○	○	○	○	○	○
API Usage Faults from incorrect use of server-side APIs, like malformed requests or missing parameters (excluding tool or client APIs)	○	○	○	○	○	○	○	○
Server Dependency Faults from missing, outdated, or misinstalled dependencies and incompatible library versions	○	○	○	○	○	○	○	○
Server Platform Incompatibility Faults related to the MCP server's functionality on different platforms (e.g., OS, Docker).	○	○	○	○	○	○	○	○
Server Infrastructure Server functionality errors caused by hardware or infrastructure faults(local or cloud)	○	○	○	○	○	○	○	○
Server Logging Faults with MCP server logging, such as wrong log levels, missing logs, or excessive logging, excluding server–host/tool interactions.	○	○	○	○	○	○	○	○

	Encountering the fault		How severe was the fault			How much effort was required to handle the fault		
	Yes	No	Minor	Major	Critical	Low	Medium	High
Server Resource Synchronization Faults from concurrent or repeated access to shared resources, causing race conditions, file locks, or improper cleanup	○	○	○	○	○	○	○	○
Server's Resource Handling Faults caused by incorrect assumptions about files, paths, and resources, leading to conflicts and missing or inaccessible project data	○	○	○	○	○	○	○	○

Server/tool Configuration

Have you ever encountered the following faults related to **'MCP server/tool setting'**?
Faults with the MCP server interactions with tools and their components (including all entities from the tool side, including tool, resource, and prompt*)*

	Encountering the bug		How severe was the bug			How much effort was required to handle the bug		
	Yes	No	Minor	Major	Critical	Low	Medium	High
Tool Authentication Faults in the authentication mechanism (e.g., credentials or processes) when connecting to a tool	○	○	○	○	○	○	○	○
Tool Authorization Faults related to granting or revoking a tool's permissions to access resources or perform actions	○	○	○	○	○	○	○	○
Tool Call/execution Faults related to the code of the tools defined in the server or calling the tools outside the server (through a connection), invoking the tools using correct parameters, etc	○	○	○	○	○	○	○	○
Tool Response Handling Faults in handling tool responses, including processing, validating, and formatting them to be returned to the host	○	○	○	○	○	○	○	○
Tool Discovery & Registration Faults with the tool descriptions the MCP server provides, tools' exposure to the host, and proper registration of tools on the server	○	○	○	○	○	○	○	○
Tool Dependency Faults from missing or misconfigured dependencies required by tools, independent of the server's dependencies	○	○	○	○	○	○	○	○
Tool Connection Synchronization Faults with timing and synchronization between the MCP server and third-party tools, including timeouts	○	○	○	○	○	○	○	○

	Encountering the bug		How severe was the bug			How much effort was required to handle the bug		
	Yes	No	Minor	Major	Critical	Low	Medium	High
Incorrect Tool Connection Setting Faults from incorrect connection settings between the server and third-party tools, excluding protocol and timing issues	○	○	○	○	○	○	○	○

Server/host Configuration

Have you ever encountered the following faults related to '**MCP server/host setting**'?
Faults in interactions between the MCP server and host clients, including IDEs, applications, and UI shells

	Encountering the bug		How severe was the bug			How much effort was required to handle the bug		
	Yes	No	Minor	Major	Critical	Low	Medium	High
Host's Server Config Faults in setting up server configuration located inside the MCP host (mostly JSON file)	○	○	○	○	○	○	○	○
Host Logging Faults in handling server-to-host logs, including log levels and related settings	○	○	○	○	○	○	○	○

	Encountering the bug		How severe was the bug			How much effort was required to handle the bug		
	Yes	No	Minor	Major	Critical	Low	Medium	High
Large Language Model (LLM) Faults in managing MCP host interactions with LLMs, including prompt handling and correctness of server selection or tool call instructions	○	○	○	○	○	○	○	○
Host Dependency Faults from missing, incorrectly installed, or incompatible library versions required by the MCP host environment	○	○	○	○	○	○	○	○
Host's Hook Config Faults with setting up and configuring host lifecycle hooks or event-triggered actions	○	○	○	○	○	○	○	○
Host License Faults with server permissions to access resources or perform actions on the host	○	○	○	○	○	○	○	○
Host Connection Configuration Mismatch Faults from incompatible protocols used in client-server connections (e.g., stdio, SSE, POST/GET)	○	○	○	○	○	○	○	○
Incorrect Host Connection Setting Faults from incorrect server-to-host connection settings, excluding protocol and timing issues	○	○	○	○	○	○	○	○
Host Connection Synchronization Faults from delays, timeouts, or asynchronous messaging in server-host communication	○	○	○	○	○	○	○	○
Session Isolation Faults with session isolation, leading to data or resource leaks	○	○	○	○	○	○	○	○

	Encountering the bug		How severe was the bug			How much effort was required to handle the bug		
	Yes	No	Minor	Major	Critical	Low	Medium	High
Session Update Faults with updating or synchronizing host and server sessions in multi-session environments	○	○	○	○	○	○	○	○

Documentation

Have you ever encountered the following faults related to **'MCP server documentation'**?

	Encountering the bug		How severe was the bug			How much effort was required to handle the bug		
	Yes	No	Minor	Major	Critical	Low	Medium	High
Documentation Faults with formal documentation of the MCP server, including README files, tutorials, or examples	○	○	○	○	○	○	○	○

Have you ever encountered any fault in development of MCP servers that have not been mentioned in this survey?

If yes, could you please describe them?

Any other comments:

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